

Assignment 4: Bias-Variance Decomposition and Stochastic Optimization

Víctor Manuel Ruiz Pereda

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1 Analysis of the Bias-Variance Generalization Tradeoff

In statistical learning theory, the systemic generalization error of a predictive mapping model f optimization can be decomposed using the following additive mathematical theorem:

$$\text{MSE}(x) = \sigma^2 + \left(f(x) - \mathbb{E}[\hat{f}(x)]\right)^2 + \mathbb{E} \left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)]\right)^2 \right]$$

This foundational relation establishes that total predictive risk is bounded by the intrinsic irreducible environmental noise (σ^2), the squared structural approximation bias, and the variance of the model space parameterization.

Based on the computational experiments executed via `scikit-learn` mapping cosine wave functions, we evaluate the parametric boundaries:

- **Underfitted Regime (High Bias):** Restricting the model mapping space to low-degree configurations prevents the architecture from mapping the non-linear curvature, resulting in high functional bias.
- **Overfitted Regime (High Variance):** Excessive dimensional polynomial expansions track the random variations of the dataset. The validation space error increases because the function minimizes training residuals by losing global generalization capabilities.

2 Stochastic Gradient Descent (SGD) Dynamics

Stochastic Gradient Descent accelerates optimization convergence trajectories by computing empirical loss functions gradients on local subsets or singular randomized instances (x_t, y_t) rather than evaluating full historical data limits:

$$w_{t+1} = w_t - \gamma \nabla_w \mathcal{E}(w_t; x_t, y_t)$$

The iterative injection of localized variance introduces a stochastic path in the optimization profile. Rather than decreasing monotonically, the empirical loss fluctuates. This algorithmic noise allows the operational model to escape shallow non-convex local extrema or unstable saddle points, tracking towards robust parameter configurations with superior validation profiles.